

SIMATS ENGINEERING

ASSESSMENT TOOL 3: Mock Interview

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO5: *Design big data processing systems using the Hadoop ecosystem including HDFS, MapReduce, and YARN.* (BL5)

Assessment Tool Weightage: 10%

Dave's Taxonomy: Articulation (Level 4)
Question Count: 10

Total Marks: 25

Code of Conduct

I **H.Mohammed Nawaz (192325060)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student : **H.Mohammed Nawaz (192325060)**

Assessment Tasks Mock Interview

Answer the following interview-oriented questions clearly and concisely. Support your responses with architecture-level reasoning wherever appropriate.

1. Explain the difference between HDFS and traditional file systems.
2. Why is replication important in distributed storage systems?
3. How does MapReduce divide work between map and reduce phases?
4. What role does YARN play in large-scale Hadoop processing?
5. Differentiate NAS and SAN in an enterprise data environment.
6. What is the purpose of ETL in a big data pipeline?
7. How does Apache NiFi help in data integration workflows?
8. What are the key failure points in a Hadoop cluster and how would you handle them?
9. How would you design a secure data ingestion pipeline for a large organization?
10. How would you explain a scalable Hadoop-based processing system to a non-technical manager?

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Technical Knowledge	25%	Shows deep and accurate technical understanding	Good technical understanding	Basic understanding	Weak understanding
Problem Solving	20%	Responds with strong architecture-level	Reasonable reasoning	Basic reasoning	Weak reasoning

1. Difference between HDFS and traditional file systems

HDFS is a distributed file system designed for storing very large files across multiple computers, while traditional file systems usually store data on a single machine or storage device. HDFS provides replication, fault tolerance, and scalability using multiple DataNodes.

2. Why is replication important in distributed storage?

Replication keeps multiple copies of data on different nodes. If one node fails, another copy can be used, preventing data loss and improving availability. In HDFS, a replication factor of **3** is commonly used for fault tolerance.

3. How does MapReduce divide work between map and reduce phases?

The Map phase divides the input data into smaller parts and processes them in parallel to produce key-value pairs. The Shuffle and Sort phase groups similar keys, and the Reduce phase combines these values to produce the final result.

4. What role does YARN play in large-scale Hadoop processing?

YARN manages cluster resources such as CPU and memory and allocates them to different applications. It allows multiple jobs such as MapReduce, Spark, and machine-learning workloads to run on the same Hadoop cluster.

5. Differentiate NAS and SAN in an enterprise data environment

NAS provides file-level storage over a network and is commonly used for shared files. SAN provides block-level storage and appears to servers like local disks. SAN generally offers high performance, while NAS is simpler for file sharing.

6. What is the purpose of ETL in a big data pipeline?

ETL means Extract, Transform, and Load. It extracts data from different sources, cleans and transforms it into a useful format, and loads it into a storage or analytics system. It improves data quality and consistency.

7. How does Apache NiFi help in data integration workflows?

Apache NiFi automates the movement and transformation of data between different systems. It provides visual data flows, routing, filtering, monitoring, and error handling, making it easier to build and manage data ingestion pipelines.

8. What are the key failure points in a Hadoop cluster and how would you handle them?

Important failure points include NameNode, DataNodes, network, disks, and resource managers. I would use NameNode High Availability, data replication, rack awareness, monitoring, automatic failover, and regular backups to handle these failures.

9. How would you design a secure data ingestion pipeline for a large organization?

I would use NiFi or an ETL tool → secure transfer → validation → HDFS/data lake → processing. I would apply encryption during transfer and storage, authentication, authorization, access control, auditing, and data validation to protect sensitive information.

10. How would you explain a scalable Hadoop-based processing system to a non-technical manager?

I would explain it as many computers working together like a team. Instead of one computer storing and processing all the data, Hadoop distributes the work across multiple machines, so the organization can handle increasing data by simply adding more machines

Mock Interview Rubric

		reasoning			
Communication	20%	Answers are confident, precise, and professional	Mostly clear communication	Moderate clarity	Weak communication
Practical Awareness	20%	Connects concepts to realworld implementation well	Some practical relevance	Limited practical relevance	No practical linkage
Confidence and Professionalism	15%	Highly confident and professional	Generally professional	Moderately professional	Low confidence and professionalism